INCH-POUND
MIL-DTL-44132C
23 AUGUST 2000
SUPERSEDING
MIL-V-44132B
20 FEBRUARY 1990

DETAIL SPECIFICATION

VEST, CONDITIONING MICROCLIMATE: AIR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers an aramid knitted cloth vest with body conditioned air distribution accessories.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center Philadelphia, Clothing & Textiles Directorate, ATTN: DSCP-COET, 700 Robbins Ave., Philadelphia PA 19111-5092 by using the self- addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document.

AMSC N/A FSC 8415

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SPECIFICATIONS

FEDERAL

DDD-L-20 - Label; For Clothing, Equipage, and Tentage, (General Use)
A-A-55195 - The=read, Para-Aramid, Spun, Intermediate Modulus

STANDARDS

MIL ITARY

MIL-STD-129 - Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg.4D, 700 Robbins. Avenue, Philadelphia, PA 19111-5092.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

2-11-1 - Vest and Connector Assembly, Conditioning Microclimate Air

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-EMSS, Natick, MA 01760-50114.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging
D 6193 – Standard Practice for Stitches and Seams
D 5118 – Standard Practice for Fabrication of Fiberboard Shipping Boxes
D 1974 – Standard Practice for Method of Closing, Sealing and Reinforcing Fiberboard Shipping Containers.
D 6199 - Standard Practice for Quality of Wood Members of Containers and Pallets.

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(Application for copies should be addressed to the American Society for Testing and materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428)

COLOR ASSOCIATION OF THE UNITED STATES

Department of Defense Standard Color Card for Sewing Thread

(Color cards may be available from the Color Association of the United States, 343 Lexington Avenue, New York, NY 10016-0927. If color cards are not available from the Color Association, individual color samples may be obtained from the contracting activity or as directed by the contracting activity.)

AMERICAN NATIONAL STANDARDS INSTITUE (ANSI)

ANSI/ASQC Z1.4 – Sampling Procedures and Tables for Inspection by Attributes

(Applications for copies should be addressed to the American Society for Quality Control, 310 West Wisconsin Avenue, Milwaukee, Wisconsin 53203).

(Non government standards and other publications are normally available from the organizations that prepare or distribute the documents. These document also may be available in or through libraries or other informational services.)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Samples. Samples of the end item, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this specification may appear in the sample, in which case this specification shall govern.

3.3 Materials. The materials shall be as specified on Drawing 2-11-1 and all subsidiary drawings and parts lists pertaining thereto. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

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3.3.1 Thread, para-aramid. The thread for all stitching shall be size T-50, 3-ply, and shall be dyed Camouflage Green 483, C.A. 34094**,** conforming to A-A-55195.

3.4 Design and construction. The design and construction of the vest shall be as specified herein, and as shown on Drawing 2-11-1 and all subsidiary drawings and parts lists pertaining thereto.

3.4.1 Stitches, seams and stitchings. All stitches, seams, and stitching shall be performed using stitch type 301 of ASTM D 6193 with and 8 to 10 stitches per inch and 1/8 +/- 1/16 inch stitching margin.

3.4.1.1 Automatic stitching. Automatic stitching machines may be used to perform any of the of the required stitch patterns provided the requirements for the stitch pattern, stitches per inch, and size and type of thread are met, and at least three or more tying, overlapping, or back stitches are used to secure the ends of stitching.

3.4.1.2 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1 inch (1/2 inch for boxstitching), except where ends are turned under or caught in other seams or stitching. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

3.4.1.2.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during stitching, except prestitching, the stitching shall be repaired by restarting the stitching a minimum of 1 inch (1/2 inch for boxstitching), back of the end of the stitching.1/

b. Except for pre-stitching, thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of 1 inch back of the defective area (1/2 inch on boxstitching), continue over the defective area, and continue onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials, and re-stitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.4.2 Thread ends. All thread ends shall be trimmed to ¼ inch maximum length.

3.4.3 Setting of eyelets. Holes shall be pre-punched to receive the eyelets and shall be smaller than the outside diameter of the eyelet barrel so that the barrel must be forced through the hole.

3.5 Labels. Each vest shall have an identification and instruction label conforming to type VI of DDD-L-20. The labels shall show fastness to laundering.

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3.5.1 Identification label. The identification label shall conform to class 1 and shall contain the following information:

VEST, CONDITIONING MICROCLIMATE: AIR
CONTRACT NUMBER
NATIONAL STOCK NUMBER
MANUFACTURER

3.5.2 Instruction label. The instruction label shall conform to class 3 and shall contain the following information:

VEST, CONDITIONING MICROCLIMATE: AIR

(1) Vest with manifold attached is worn over undershirt. Close front zipper and adjust side straps for snug fit. Hose and connector are to exit out from under other clothing layers, as required, either from right or left side.
(2) Attaching manifold to vest:
    (a). Attach round manifold with hook and pile fastener tape to center of back side of vest.
    (b). Attach front manifold with hook and pile fastener tape below slide fastener.
    (c). Secure all hoses to vest with hook and pile fastener tape straps.
(3) Vest without its manifold assembly may be hand washed using warm water with soap or detergent. To dry, hang and allow to air dry.

DO NOT REMOVE THIS LABEL

3.5.3 Bar Code Label/Tag. Each item shall be individually Bar-coded with a Type VIII, class 17, Label/Tag of DDD-L-20. The Label/Tag shall be located so that it is completely visible on the item when it is folded and/or packaged as specified and shall cause no damage to the item.

3.6 Workmanship. The vest shall conform to the quality of product established by this specification.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Quality conformance inspection (see 4.3).

4.2 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4 and 4.4.1. Any nonconformance shall be cause for rejection of the first article.

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4.3 Conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with ANSI/ASQC Z1.4.

4.3.1 Examination. Each vest and hose assembly shall be examined for compliance with the requirements specified in section 3 and subsequent government drawings. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.
In addition, the first production lot or each of the components listed below shall be examined for conformance to all dimensions specified on the referenced drawings. Any dimension exceeding the applicable tolerance specified on the applicable drawing shall be classified as a defect. The lot sizes shall be expressed in units of components. The sample unit shall be one component. For each component lot, the inspection level shall be S-1 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5. Failure of any lot shall be cause for correction of the mold or process and for examination of the next produced lot from that mold or process. Any subsequent lot in which a change in materials, mold, or process has occurred shall be examined as above. If there are no changes, subsequent lots shall be accepted on the basis of a certificate of compliance certifying that there have been no changes in materials, mold, or manufacturing process.

Component	Drawing
Connector assembly 1/	2-11-11 2-11-12 2-11-13 2-11-14
Back air distribution manifold	2-11-24
Front air distribution manifold	2-11-25

1/ Component must be disassembled prior to performing dimensional examination.

4.4 In-process inspection. Inspection of subassemblies shall be made to ascertain that construction details which cannot be examined in the finished products are in accordance with specified requirements. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance. Whenever nonconformance is noted, corrections shall be made to the parts affected and lot in process. Components that cannot be corrected shall be removed from production.

4.4.1 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of vests. The sample unit shall be one completely fabricated vest. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for total (major and minor combined) defects.

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TABLE I. End item visual defects

Examine	Defect	Classification	
		Major	Minor
Rows of stitching	Any row of stitching missing	101	
Open seams	½ inch or less		201
	More than ½ inch	102	
Raw edge (on edge required to be finished)	More than ½ inch when securely caught in stitching		202
NOTE: Raw edge not securely caught in stitching shall be classified as an open seam.			
Seam and stitch type	Wrong seam or stitch type	103	
Stitch tension	Stitch tension loose, resulting in loose bobbin or top thread.		203
	Stitch tension excessively tight, resulting in puckering of material.		204
Stitches per inch	Three or more stitches less than minimum specified.	104	
	Up to two stitches less than minimum specified.		205
	One or more stitches in excess of maximum specified.		206
Stitching Ends	Not secured as specified		207
Thread Breaks, skipped stitches, or runoffs	Not overstitched as specified.		208
NOTE: Thread breaks or two or more consecutive skipped or runoff stitches not overstitched shall be classified as an open seam.			

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification	
		Major	Minor
Stitch margin or gage	Not within specified tolerance		209
Fabric Defects	Any cut, tear, break or mend	105	
	Aabraision mark, resulting in a weak area.	106	
	Buckled, ridgy or wavy (olefin spacer cloth)		210
	Not firmly or tightly woven, edges frayed or scalloped (webbing and tape).		211
Slide Fastener	Not functioning properly, failing to effect a secure closure or to open freely.	107	
	Not specified type or size.	108	
	Slider jams or fails to interlock.	109	
	Fasterner tape cut or torn.	110	
Fastener Tape	Any hole, cut or tear.	111	
	Hooks flattened, broken, or missing, impairing function.	112	
	Width not as specified.	113	
	Length greater or less than specified on parts list by:		
	a. More than 3/8 inch	114	
	b. 1/8 to 3/8 inch inclusive		212
Plastic components	Any rough or sharp edge, or not free of flash	115	
	Wrinkle, pimple, or crater	116	
	Crack, break, rupture, or fracture	117	
	Bent, twisted, malformed, or otherwise impaired	118	
	Design detail deviating from drawing requirements (unless otherwise classified herein)	119	
	Any joint not securely bonded	120	
	Color not as specified		213
	Any adhesive/solvent stains or scratches		214

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification	
		Major	Minor
Hose Assemblies	Distance between reinforcement more or less than specified:		
	By 1/8 inch		215
	By 1/4 inch	121	
	Eyelets:		
	One or more missing or not securely attached through fabric and small air distribution hose.	122	
	Deviating in position from required angle to the extent that intended function will be seriously impaired or not performed.	123	
NOTE: This defect should be observed with the hose manifold assembly removed from the vest. The front and the rear manifolds shall be held firmly together face to face or back to back. The ends of small hoses shall be parallel and adjacent where affixed to manifolds. With no force being applied to the hoses, the loop between the front and back manifolds shall not be skewed in any direction beyond the outside edge of the front manifold.			
	Hose assemblies not a tight fit to manifolds(i.e., tunnels, creases, or puckers in bonded area)	124	
	Any insecure bond of hose assemblies to manifolds	125	
	Heat shrinkable sleeving:		
	Not securely affixed.		216
	Not fully covering raw edge of cloth		217
Thread ends	Any length extending beyond heat shrinkable sleeving	126	
	More than 1/2 inch in length	127	
	1/4 to 1/2 inch in length		218
Cleanness	Any spot or stain		219
Labels Identification and instruction	Omitted, incorrect, illegible, not as specified, or not positioned as specified.		220

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification	
		Major	Minor
Bar Code Label/Tag	Bar code omitted or not readible by scanner.		221
	Human readible interpretation (HRI) omitted		222
	Not visible on folded, packaged item.		223
	Causes damage to the item.	128	

4.4.4 End item dimensional examination. The end item hose assemblies shall be examined for the defects listed in table II. For this examination, the hose assemblies shall be removed from the sample vests and placed on a flat surface at full length with the hook tape fastener on the manifolds facing up. The lot size shall be expressed in units of vests. The sample unit shall be one complete vest. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE II. End item dimensional defects

Examine	Defect	Classification	
		Major	Minor
Large air distribution hose	Length more than ¼ inch more or less than specified tolerance	101	
	Length up to 1/14 inch more or less than specified tolerance		201
Small air distribution hose	Length more than 1/8 inch more or less than specified tolerance	102	
	Length up to 1 /8 inch more or less than specified tolerance		202
Cloth covered foam	Length more than 1/8 inch more or less than specified tolerance (from edge of sleeve to edge of sleeve)	103	
	Length up to 1/8 inch more or less than specified tolerance (from edge of sleeve to edge of sleeve)		203
	Not positioned from back manifold as specified		
	More than 1/8 inch	104	
	Up to 1/8 inch		204

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TABLE II. End item dimensional defects (con't)

Examine	Defect	Classification	
		Major	Minor
Eyelet vents	Not positioned from edge of sleeve as specified: More than 1/8 inch Up to 1/8 inch	105	 205

4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling. Bulged or distorted container
Content	Number of vests per container is more or less than required Number of vests per unit pack is more or less than specified

4.4.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1i and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

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Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

5. PACKAGING

5.1 Packaging. The packaging requirements shall be as specified in the contract or purchadse order. (see 6.2)

5. 3 Palletization. Palletization is required as per contract or purchase order.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The microclimate air conditioning vest, consisting of a vest with attached air distribution accessories, is designed to distribute conditioned air over the wearer's torso. It is intended to be worn under a chemical protective garment in M1A1 US Army Tanks.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Packaging (see 5.1).
g. When palletization is required (see 5.3).

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6.3 First article. When a first article is required, it will be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a pre-production sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to samples, address the contracting activity issuing the invitation for bids or request for proposals.

6.5 Subject term (key word) listing.

Clothing
Cooling
Protective garment

Custodians:
Army - GL
Navy/MC
DLA-CT

Review activities:
Army - MD

Preparing activity:
DLA - CT

(Project 8415-0201)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-44132C	**2. DOCUMENT DATE (YYYYMMDD)** 20000823

3. DOCUMENT TITLE **VEST, CONDITIONING MICROCLIMATE: AIR**

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYYYMMDD)

8. PREPARING ACTIVITY

a. NAME DEFENSE SUPPLY CENTER PHILADELPHIA DPSC-COET	b. TELEPHONE *Include Area Code)* (1) Commercial (215) 737-2405 (2) AUTOVON 444-2405
c. ADDRESS *(Include Zip Code)* 700 ROBBINS AVENUE (Bldg. 6, C & T) PHILADELPHIA, PA 19111-5092	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman road, Suite 2533, Ft. Belvoir, VA 22060-2533 Telephone (703) 767-6888 AUTOVON 427-6888

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